

APPLICATION OF CISCO PACKET TRACER 6.2 IN TEACHING OF ADVANCED COMPUTER NETWORKS

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Abstract - Cisco Packet Tracer 6.2 is simulation-based learning environment software that allows students to experiment with computer networks behaviors and helps develop their skills such as decision making, critical thinking and problem solving. This software provides a visual simulation of complex networking concepts and configurations and allows students to practice using a command-line interface. These simulation capabilities can help simplify the learning process and other visual representations of internal networking functions, such as real-time dynamic data transfers and packet content expansion. The purpose of this paper is to present and investigate the possibilities and key features of the Packet Tracer 6.2 software, important for teaching and development of modern computer networks.

I. INTRODUCTION

One thing we surely can agree on is that the Internet will indeed grow exponentially over the coming years. The deployment of applications with increase in demand for bandwidth and the development of new services, have all caused an immense advance of computer networks technology. Differentiation between the classic structures of local (LAN) and wide area networks (WAN) has become less distinct. Due to the current rate of technology innovations, a computer networks could become obsolete within two to three years. Hardware upgrades of active network devices and components to the next level of technical development can usually be implemented with more or less effort.

There is high demand for computer networking skills in industry, education, military, commerce, wireless/wired communication technologies etc. Consequently, computer networking courses are becoming very popular in universities, higher technical schools and private training institutions across the globe [1]. However, it is often difficult to motivate students to learn basic computer network principles and technologies, because students find the topics rather abstract when they are presented using a traditional lecture format. One of the main imperfections about network devices is that students

often cannot view or access the real network equipment and cannot analyse the data packages coming from the various real network devices [2].

An alternative way is to provide students with a network simulation software. While simulators cannot provide students with some practical skills such as cabling or physical devices connectivity, they represent a useful and cost-effective approach to understand concepts of computer networks, protocols and applications better than traditional tools do.

There are many software tools for network protocol analysis, both in commercial and open source products. Wireshark [3] is very easy to use and free network protocol analyser. However, one of the disadvantages is that it doesn't provide visual connection between OSI model protocols of each network layer. Commercial NetSim and free GNS3 [4,5] software provide visual virtual network devices, so that they solve some problems of real network devices in network teaching. However, both of them don't provide suitable visual ways for network protocol analysis.

Cisco Packet Tracer (PT) is teaching and learning virtual networking simulation software developed by Cisco Systems Inc. and widely used by the students participating in CCNA or CCNP courses offered by Cisco Networking Academy Program (CNAP) [6]. PT provides a virtual network environment with substantially details of the network operating system on individual devices. PT allows creation of realistic scenarios of various networking structures, network system configuration and network troubleshooting. PT is a graphically based (GUI) interactive software, but it also provides students with a text-based CLI (Command Line Interface), available for configuring network devices. The CLI allows students to enter partial Cisco IOS commands. Cisco IOS is network infrastructure software used on most Cisco routers and current switches [7].

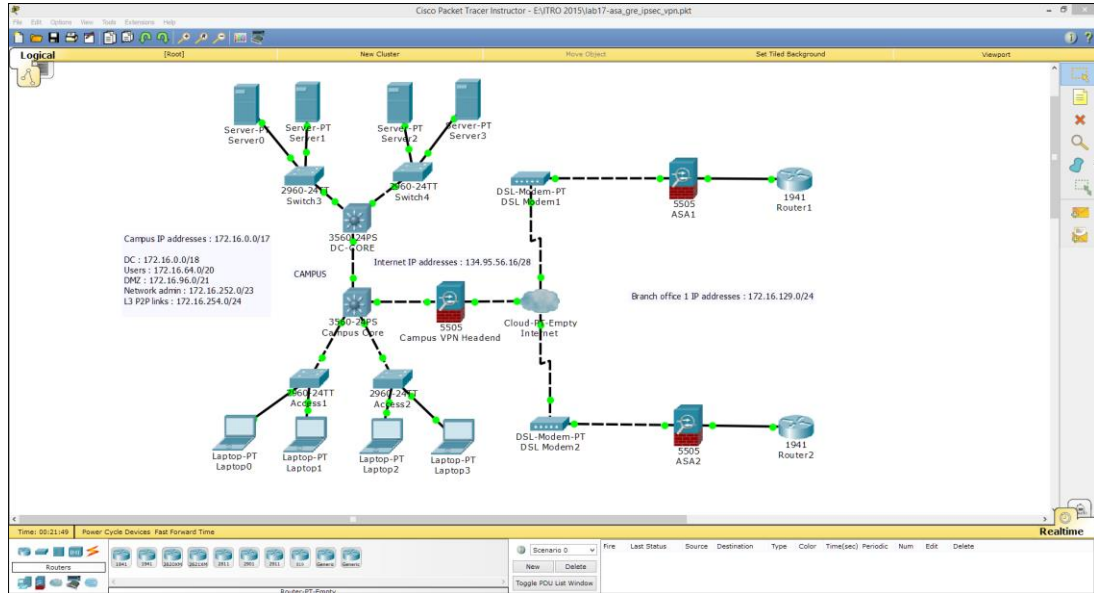


Figure 1. Packet Tracer network devices topology map

II. FEATURES OF PACKET TRACER 6.2

A. New devices and protocols

In February 2015 Packet Tracer 6.2 was released in two versions - Student and Instructor. PT 6.2 includes an ASA 5505 firewall with CLI configuration (but no ASDM or CCP tools) [8,9]. It also includes a netflow collector as a desktop application in the server device, routing protocols for IPv6 (OSPFv3, EIGRPv6, RIPng), DHCP snooping, IPv6 CEF and IPsec commands. This PT version includes a new Cisco 819 ISR router with an embedded wireless access point and some new devices: 3G/4G cell tower, Central Office (CO) server and 3G/4G support for end devices (smartphone, tablet, server etc.), [Table 1].

B. Workspaces and operating modes

PT has two types of workspace: logical and physical. Logical workspace allows students to build logical network topologies by placing, connecting,

and clustering virtual devices. The physical workspace provides a graphical visualization of the logical network and represents how network devices would look in a real environment. The physical view enables geographic representations of networks, including multiple cities, buildings and wiring closets.

There are two operating modes to visual representation of a network behavior: real-time and simulation mode. The real-time mode enables students to gain configuration practice because the devices in network look and behave exactly the same as real Cisco devices. In simulation mode they can see, control and analyse time intervals and propagation of data across a network, and can learn how to troubleshoot network failures. This significantly helps most of them understand the fundamental concepts behind network operations. In Figure 1 the example of the PT network devices topology map is shown. This topology represents how to create site-to-site IPsec VPN using ASA 5505 firewall. A small branch office is securely connected to the enterprise campus over the internet using a broadband DSL modem connection [10].

C. Modular network devices

Graphical representations visually simulate network equipment, and offer the ability to insert interface cards into Cisco modular routers and switches. Selecting switches or routers from the device-type selection box lists both Cisco devices and some devices labeled Generic. These are custom PT devices running on Cisco IOS, but the slots that hold the modules are different.

TABLE 1. PACKET TRACER 6.2 - NEW FEATURES

HTTP server (server device)
– Javascript and CSS support – HTTP and FTP linked file management support – External file import inside Packet Tracer HTTP server
New devices
– Cisco 819 Integrated Service Router with WLAN access point capability and 3G/4G connectivity – 3G/4G telephony cell tower – 3G/4G central office (CO) server with coaxial connectivity with up to 8 cell towers – Wired sniffer with repeater functionality

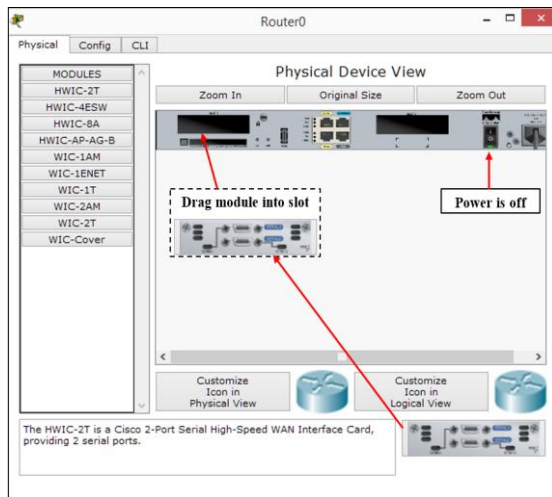


Figure 2. Setup of Cisco router with HWIC-2T module

A device module is a piece of hardware containing several device interfaces. For example, a HWIC-2T module contains two Serial High-Speed WAN ports. Similar to a real Cisco router, the virtual device has to be powered off in order to add or remove modules (Figure 2). The power switch is on the right-hand side of router, and click on this switch will turn it off. To add a module, drag it from the modules list and drop onto an empty slot. To remove a module, power off the router and drag it from the slot back to the module list.

PT is supported by a various networking devices which can be used to create different networking scenarios. Except routers and switches, they include hubs, wireless and wired end devices, WAN emulation, custom made devices, multi-user connections, PCs, laptops, servers, printers, IP phones, VoIP devices, analog phones, TVs, tablets, ASA 5505, CO servers, Cell towers etc.

D. Device connections and settings

Various types of cables which can be used to connect various networking devices in a PT are: console cable, copper straight-through/cross-over cable, coaxial, phone, fibre cable, Serial DTE/DCE and octal cable. After connecting devices, students can see a light at each end of the cable. This indicates the state of the connection, as follows: Bright green (physical link is up, but it doesn't indicate the status of the line protocol), Blinking green (link activity), Red (physical link is down and it can be caused by incorrect cables or by a port being administratively shut down), Amber (this appears only on switches, and indicates that the port is running the STP Spanning Tree Protocol algorithm to detect layer 2 networking loops) [11].

For the most common Cisco network devices configurations PT provides GUI setup options. The

configuration window for Cisco devices, such as routers and switches, consists of three tabs. The Physical tab is used to add or remove modules. Using the Config tab, the following can be configured: Global settings, Routing (on a router or a layer 3 switch), VLAN database (on a switch) and Interface settings (Figure 3). The CLI tab is used to configure all of the device settings supported by PT in the same manner used via the command line interface on an actual device. A student can access the CLI mode of a device either by using a terminal software when a PC is connected to a router/switch using a console cable, or by using Putty, SSH or Telnet when it is connected using a crossover Ethernet cable. Packet Tracer offers a feature to save a device that students can configure as a custom-made device, with particular set of modules.

Packet Tracer 6.2 has some improvements about device clustering. A cluster is a feature of the logical workspace and hence does not affect how devices are displayed in the physical PT workspace. Because when large network topologies are created, sometimes it becomes difficult to understand them after a while. By combining several devices into a single cloud icon, students can obtain access into specific clusters and network subnets.

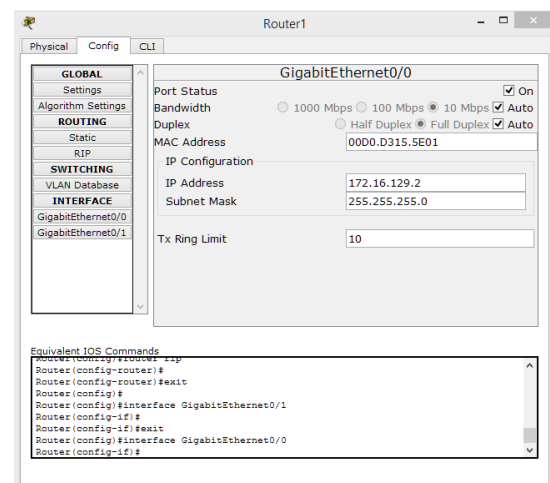


Figure 3. Configuration tab options for Cisco router

III. SIMULATION MODE AND ANALYSIS

After creating a network topology, connectivity between devices can be tested by using either simple or complex PDU (Packet Data Unit). The PDU information window allows students to open a packet and look inside to see how it is being processed at each layer of the OSI Model. It is possible to do the same testing by pinging devices from their CLI interface, but using the PDU option is quicker for large network topologies. There are two options for PDU connectivity testing: Simple

and Complex PDU. The Simple PDU uses only ICMP (Internet Control Message Protocol). Creating a Complex PDU allows student to control parameters of the packet such as: Protocol, Source and Destination IP, Port, TTL and Sequence number and also allows to test connectivity to specific interfaces on a device.

Using simulation mode, students can see and analyse packets step by step flowing from one network node to another, and can also see detailed packet information categorized by OSI layers. In this mode PT allows pause the simulation, or step forward/backward in time, and investigate many types of information on specific objects at specific times.

PT uses the Realtime/Simulation tab to switch to the simulation mode. Simulation mode has the Auto Capture/Play button to begin packet capture (Figure 4). Pressing this button all of the network traffic (chosen under event filters) being continuously captured until this button is pressed again. Capture/Forward button represents manual mode of the previous button and has to be pressed each time to move the packet from one place to another. Button Back moves the process one step back each time it is clicked on. Student can try a Simple PDU and the event list will be populated with three entries, indicating the creation of an ICMP packet, echo sent, and ICMP reply received. A timer at the bottom of the Event List window shows the total time that has elapsed since the beginning of the simulation. Constant Delay can be turned off so that actual processing delay and propagation delay is added to the simulation [12].

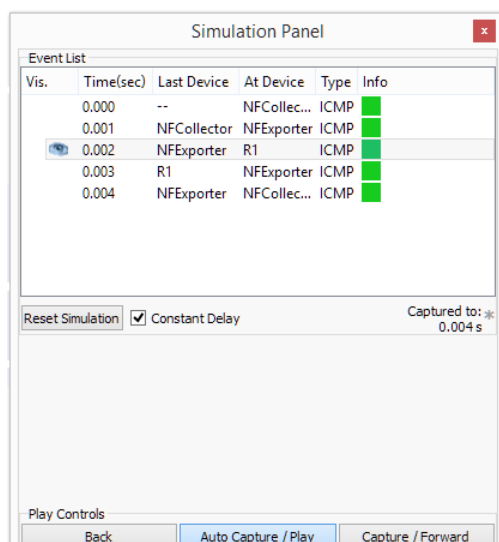


Figure 4. Event List is a part of Simulation Panel

Student can either click on the Info box for a specific PDU, or click on the PDU in the topology window to bring up the PDU Information window. The OSI Model shows the de-encapsulation and encapsulation process. Clicking on a layer will display detailed information about decisions made at that layer. The students can use OSI model tab to:

- demonstrate how switches process only to OSI Layer 2 and routers process to Layer 3
- show encapsulation/decapsulation to accommodate different interfaces
- show operation of an ACLs (Access Control Lists)
- show operation of NAT (Network Address Translation)
- show what happens to a packet with no ARP table entry
- show routing decisions about the packet either forwarding it or dropping based on routing table entries

The data for different OSI layers and the formats of network protocols can be viewed as shown in Figure 5.

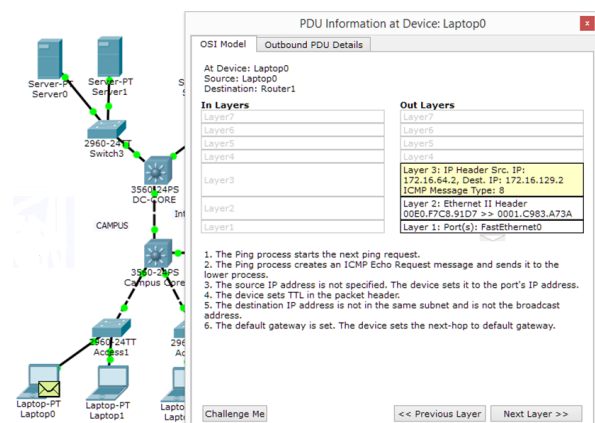


Figure 5. Layer 3 of OSI networking model

Students can quiz themselves on the encapsulation process by entering Challenge Mode. The Challenge Me button from the OSI Model tab of the PDU Information window starts Challenge Mode (Figure 5). The layer details are hidden, and the information window is replaced by a question window that asks what the device does to a PDU on a given layer. Students select from a multiple-choice list. If they answer correctly, the details for that layer are shown and the question window advances to the next layer. The Hint button provides hints.

IV. ADVANCED PACKET TRACER OPTIONS

Multiuser peer-to-peer mode of Packet Tracer 6.2 allows collaborative construction of virtual networks over a real network. The multiuser feature enables collaborative and competitive interactions, providing the option to progress from individual to social learning and features opportunities for collaboration, competition, remote instructor-student interactions, social networking and gaming [13]. This new feature has opened the door to develop many interesting new activities such as interactive and dynamic troubleshooting and serious gaming for introductory networking classes. PT multi-user activities can make networking more interesting to learn and lead to greater student engagement. The multiuser capability allows connection of remote instances of PT on separate machines [14].

The Activity Wizard guides students through the creation of an assessment. It is made accessible by navigating to Extensions>Activity Wizard. The Activity Wizard allows students to author their own learning activities by setting up scenarios using instructional text, and creating initial and final network topologies and predefined packets. Instructions in Activity Wizard are built in simple HTML format and also includes grading and feedback capabilities [15].

The physical workspace is divided into four layers to reflect the physical scale of real life environments: Intercity, City, Building and Wiring Closet. The Wiring Closet is the final layer that contains devices placed in the logical topology and it doesn't have any specified area (Figure 6).

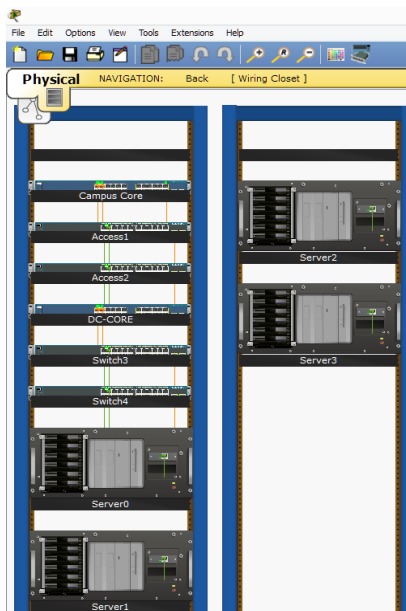


Figure 6. The physical workspace (Wiring Closet) with graphical view of network devices and connections

Packet Tracer 7.0 is under development and could include new IoE (Internet Of Everything) items and an IoE registration server. Last known build is Packet Tracer 7.0 build 70 as of 2.3.2015.

V. CONCLUSIONS

There are various benefits and advantages of using a Packet Tracer in learning basic and advanced concepts of computer networks. Because computer networks can be difficult to understand theoretically, PT has lots of features to create various scenario based labs. After doing more practice on a PT networking scenarios, students gain lot of confidence to work on real-time networking devices.

In this paper, we also presented some additional tools that can significantly help students to become more familiar with advanced computer network topologies.

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